Ground Truth Under Attack: Defending In-Situ Earth-Observation Data Against Valid-Signature False-Data Injection

Anonymous Phase 1 Submission
TSYP14 Technical Challenge, Track 2: False-Data Injection

***Abstract*—A digital signature proves who sent a sensor reading, not that the reading is true. We present a four-layer defensive prototype for the in-situ sensing segment of the Earth-observation chain. Layer 1 makes in-transit and stored-record tampering, replay, deletion and unknown senders deterministically visible with Ed25519 signatures, per-stream sequence numbers and hash chains over mutual-TLS MQTT. Layers 2 and 3 address the harder case of correctly signed false values: each reading is compared with its own past, two disjoint rings of neighbours and physical rules, and an Extra Trees classifier per variable turns 37 causal features into a calibrated p-value and an alert. Layer 4 converts provenance verdicts and alerts into a trust state per sensor, withholds or imputes suspicious values, and logs every decision in a hash-chained audit trail. On a full year of real WegenerNet data (2024) with 12 attack families, the frozen detector raises at least one alert on 79–82 % of attacks, while flagging 32–35 % of the individual manipulated readings, with 57–60 % precision and 0.013 false-alert episodes per stream-day; it doubles F1 over a rule ensemble built from the same signals. A live 30-minute tick for 486 streams takes 0.2 s on a 2-core CPU with 0.85 GB peak memory; Ed25519 provenance adds 104 bytes per record.**

***Index Terms*—false-data injection, data provenance, anomaly detection, in-situ sensing, Earth observation, Ed25519**

I. Introduction and Segment Analysis

Ground stations measuring temperature, relative humidity and precipitation feed frost and drought alerts, irrigation, parametric insurance and the validation of satellite products. A corrupted observation does not crash the pipeline: it flows on as a plausible number. Stations are physically exposed, run on low-cost hardware and are often reached over shared networks, which makes both failure modes of the challenge realistic: false-data injection and silent failure.

We defend the *in-situ sensing* segment (Track 2) on two axes. *Provenance* establishes who sent a record and that it was not altered, replayed or dropped. *Plausibility* establishes whether a correctly signed value is consistent with history, neighbours and physics. Neither is sufficient alone: a compromised or physically spoofed sensor produces perfectly valid signatures.

Our contributions are: (i) a threat model separating network/storage attackers from valid-signature attackers, with an attack taxonomy mapped to the layer expected to catch each family; (ii) a software trust anchor with fail-closed verification; (iii) a causal, calibrated detector evaluated under a strict chronological protocol with attack families withheld from training; (iv) a response layer that prevents suspicious values from reaching the analyst; and (v) latency, memory and security-overhead measurements on a laptop CPU.

II. Related Work

False-data injection was formalised for power-grid state estimation, where Liu *et al.* showed that an attacker who knows the system model can inject errors that pass residual-based bad-data checks [6]. Our setting differs: there is no physical model linking stations, so consistency must be learned from history and neighbours. Meteorological networks have long used temporal, range and spatial quality-control tests, such as the spatial regression test of Hubbard *et al.* [7]; these target sensor faults, not an adversary who can shape the error. We reuse such tests as features and learn how they combine. Unsupervised detectors such as Isolation Forest [8] and forecasting models are surveyed in [9]; in our comparison they react to rare weather and recall few attacks under a strict false-alarm budget. Our calibration (Eq. 1) is a conformal p-value [10], which makes the alert threshold a controlled tail probability on clean data. Finally, we report point metrics without point adjustment, which is known to overstate time-series anomaly detection [11].

III. Threat Model

A. Assets and trust boundaries

Protected assets are measurement values, sensor identities and credentials, freshness and order, processing evidence and detector inputs, and security decisions. Trusted components are the verifier host, the provisioned key registry, local policy and state, and the configured trust roots. The trust anchor is in software (encrypted Ed25519 keys, a local certificate authority, monotonic sequence state), which the challenge explicitly permits.

B. Attacker classes

Network/storage attacker (no sensor key): modifies, replays, drops, reorders or forges records, or edits stored readings. Expected catch: Layer 1, deterministic.

Valid-signature attacker: controls the reported values of chosen streams over a chosen interval (compromised sensor or credentials, physical spoofing). The attacker knows the current value, the causal past and training-period statistics, and may target one site, a neighbourhood of up to 6 km, or several

TABLE I
ATTACK TAXONOMY AND EXPECTED CATCHING LAYER

Category	Families	Caught by
A. Value, valid signature	bias, drift, spike, noise, flatten, freeze, invalid range, rain scale/suppress/inject, daily-cycle pattern, mimicry, value replay, coordinated patch, multivariate T+RH, network-wide bias	L2/L3: temporal, spatial, physical, cross-source
B. Record in transit	tamper value, shift timestamp, replay, drop, reorder	L1: bad signature, replay, gap
C. Stored data	rewrite stored reading, unknown sensor	L1: bad signature, unknown sensor

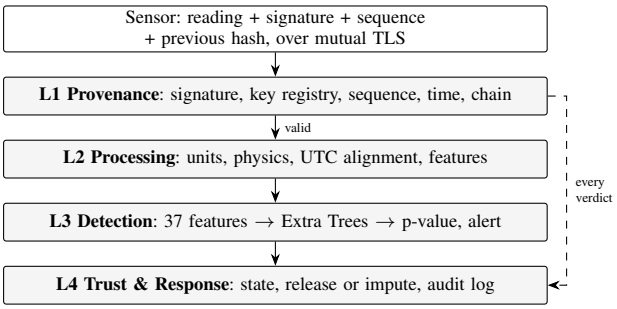


Fig. 1. The four layers run in one Python process on a laptop. Every Layer 1 verdict, including rejections, also reaches Layer 4, so failures are never silent.

variables at once. The attacker cannot alter the evaluator's labels. Expected catch: Layers 2–3, statistical, with Layer 4 limiting damage.

Out of scope: a compromised verifier host, theft of all trust anchors, unlimited denial of service, and provable separation of every network fault from an attack.

IV. Architecture and Trust Mechanism

Fig. 1 shows the pipeline. Layers exchange typed, immutable contracts with generated JSON schemas.

A. Layer 1: provenance

Each sensor holds its own Ed25519 [3] key, encrypted at rest. A reading is encoded in canonical CBOR with a per-stream sequence number and the hash of the previous record, then signed. The signer commits sequence allocation, chain advance and the exact signed bytes atomically to a durable outbox *before* publishing, so a crash never reuses a sequence number. Transport is MQTT v5 over mutual TLS only, with per-sensor topic ACLs; transport security is independent of the signature. A persistent verifier checks the signature, key-registry entry (enrolment, rotation, revocation), sequence, time window and chain, and returns an explicit status (`valid`, `bad_signature`, `replay`, `gap`, `chain_anomaly`, `unknown_sensor`, `malformed`). Authentication failure never advances state. A software anchor can be defeated by an administrator restoring an old database; no hardware anti-rollback is claimed.

B. Layer 4: trust and response

Every 30 minutes each sensor is *trusted*, *suspicious* (≥ 3 alerted ticks in 6 h) or *quarantined* (≥ 6 in 12 h, or ≥ 3 provenance failures in 2 h). Suspicious and quarantined values are imputed from the median of at least two trusted neighbours, or withheld. A single forged message never quarantines a sensor, since otherwise anyone could silence a sensor without its key; it is flagged, audited and never released. Provenance quarantine ends only by an analyst release recorded with name and reason. Thresholds were set by Monte-Carlo simulation on the detector's measured behaviour, assuming each attacked tick alerts independently with probability 0.33 (the measured reading recall): 0.072 false quarantines per sensor-month, while every simulated 24–48 h attack is quarantined (median 8 h). Real alerts cluster in time, so these figures are indicative. Decisions go to an append-only SQLite log with its own SHA-256 chain; triggers reject updates and deletions. A FastAPI dashboard shows states, raw versus released values and chain status.

V. Signal Processing

Layer 2 uses only provenance-valid readings for statistics and never modifies raw signed evidence. It normalises units (°C/°F/K with absolute-zero limits), checks physical ranges, aligns readings to exact UTC windows with explicit missingness, and emits feature windows with rolling statistics, conditional spectral features (only when sampling supports them) and neighbour comparisons. Event time and availability time are kept separate so no replay can use future data.

The detector computes 37 features per reading, each using only values up to time t (unit-tested: changing a future value never changes an earlier feature). Rain is modelled as $\log(1+x)$; scales are robust ($1.4826{\cdot}$MAD) per site and variable.

- *Temporal (11)*: residual to a forecast $b\,x_{t-1} + (1-b)\,x_{t-48}$, residual to a per-site harmonic seasonal model, a two-sided CUSUM [5] that accumulates slow drifts, 3 h and 24 h median, MAD and standard deviation, and 3 h slope.
- *Spatial (14)*: residuals to a distance-weighted median of a *near ring* (6 closest sites) and a disjoint *wider ring* (19th–24th closest), with weights capped at 45 % so one attacked neighbour cannot dominate; neighbour spread, count and co-movement.
- *Physical rules (4)*: range violation, rate of change, freeze run length (exempt for dry rain and saturated humidity) and variability ratio.
- *Context (8)*: hour and day of year, rain occurrence and amount, value and quality flag.

The disjoint wider ring answers coordinated attacks that corrupt the near neighbours; CUSUM answers drifts that never produce a single large step.

TABLE II
FROZEN DETECTOR ON 2024 (TWO ATTACK SEEDS)

	Prec.	Rec.	F1	Events	FA/sd
Seed 91000	57.1 %	31.8 %	0.408	79.2 %	0.013
Seed 191000	59.6 %	35.3 %	0.443	82.2 %	0.013
Rule ensemble	31.8 %	17–18 %	0.22–0.23	51–55 %	0.046

Rec.: recall over attacked readings. Events: share of attacks with at least one alert. FA/sd: false-alert episodes per stream-day on clean 2024 (budget 0.05).

VI. Detector Design and Evaluation

A. Model, calibration and alert rule

One Extra Trees [4] classifier per variable (120 trees, depth $\leq$ 18) learns which combinations of signals indicate manipulation; for example, a deviation from the past together with moving neighbours is weather, not an attack. It was the strongest of 11 candidates under the same false-alarm budget; GRU/TCN forecasters, a sequence autoencoder and Isolation Forest react to any unusual weather and recall few attacks under a strict budget. Training replays each 14-day episode clean and with four seeded attacked copies; a quarter of the clean samples are hard cases with large genuine neighbour residuals. Scores are calibrated on clean data as

$$p = \frac{1 + \#\{\text{clean scores} \geq s\}}{n + 1}, \qquad (1)$$

so one budget applies to all variables. Per variable, an alert is raised when $p \leq \alpha$, directly or with two-of-three confirmation, choosing the best F1 subject to ≤ 0.05 false-alert episodes per stream-day and $\geq 95\,\%$ coverage. The rule is then frozen together with the SHA-256 of every model file; the detector refuses to run if any file changes.

B. Data and protocol

We use WegenerNet Level 2 v8.0 (University of Graz, CC BY 4.0) [1], half-hourly 100 m grid, Feldbach region, 2023–2024. A 1.5 km sub-grid of cells clean for $\geq 95\,\%$ of training gives 162 sites $\times$ 3 variables $=$ 486 streams. Splits are strictly chronological: Jan–Aug 2023 training, Sep–Oct calibration, Nov–Dec rule selection, and 2024 for evaluation only. Metrics are strict point metrics without point adjustment, plus event detection (at least one alert during the attack).

C. Results

Table II reports the results. Coverage is 100 %. The gap between attacks detected (about four in five) and readings flagged (about one in three) is expected: an alert early in an attack is enough to act on, but the readings before evidence accumulates, such as the first steps of a slow drift, are missed. The false-alarm rate corresponds to about one false episode per stream every 2.5 months, four times under the budget. Against a rule ensemble fusing the same temporal, spatial and physical scores with an Isolation Forest, the learned detector roughly doubles precision, recall and F1 and produces 3.5$\times$ fewer false alarms (Fig. 2).

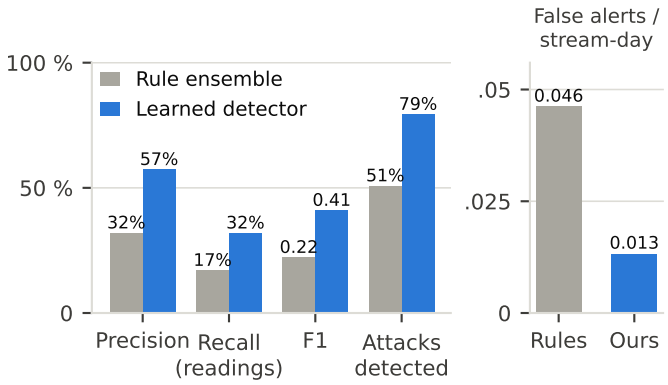


Fig. 2. Learned detector versus rule ensemble on the same signals (2024, seed 91000).

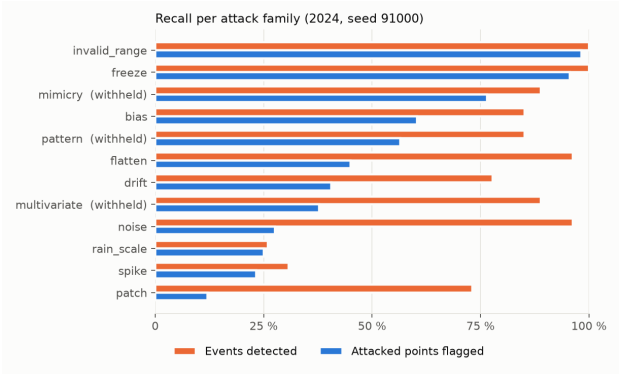


Fig. 3. Detection per attack family (2024, seed 91000). Families marked withheld were never seen in training.

Per family (Fig. 3), invalid range and freeze are always detected through the physical rules; noise and flatten (93–96 % of attacks), bias (85–93 %) and drift (78–93 %) are detected as events, though not every manipulated reading inside them.

VII. Standard Scenario and Evasion

A. Scenario

The standard scenario is versioned and seeded, and built on real public data; rebuilds are byte-identical (SHA-256 manifests) and the jury can re-run it with its own held-out attacks. It has two parts: (i) the detector protocol above, with one non-overlapping attack per family per 14-day episode, 6–48 h long, and two independent 2024 seeds; (ii) a station suite on 154 WegenerNet 5-minute stations with 16 value families plus record and storage faults, split into *dev* (108 stations), *heldout* (46 different stations, different sizes, two held-out-only shapes), *clean2024* for false alarms and *crosssource* for whole-network bias.

B. Evasion

Table III summarises how an informed attacker might evade detection and what happens.

TABLE III
EVASION STRATEGIES AND OUTCOMES

Strategy	Outcome
Unseen attack types	withheld from training: pattern 81–85 %, mimicry 89–100 %, multivariate 89–96 % detected
Different magnitudes	2024: $\pm$0.75, 1.5, 3 °C; training: $\pm$0.5, 1, 2, 4 °C (inside the range: interpolation)
Match own statistics	mimicry with training mean/variance: 89–100 % detected, spatially
Hide in the daily cycle	pattern: 81–85 % detected
Corrupt neighbours	patch up to 6 km: 69–73 % detected via the wider ring
Shift whole network[†]	neighbours blind (< 2 % alarms); cross-source baseline alarms on $\sim$100 % of cells
Stay below threshold[†]	escape sweep (rule baseline): largest size still missed per family
Forge or replay records	Layer 1 rejects 100/100 tampered records, audited

[†]Station suite, rule-based baselines (unit test and `eval/escape.py`); the trained detector is graded on it in Phase 2.

TABLE IV
COST ON A 2-CORE X86-64 CPU (NO GPU, NO CLOUD)

Measurement	Value
Live tick (486 streams, 24 h history)	median 202 ms; p95 236 ms
Share of the 30-min interval used	< 0.02 % used
Batch replay, 14 days	5.0 s (16.5 µs/reading)
Detector peak memory / model size	852 MB / 17.6 MB
Ed25519 sign / verify (incl. chain)	17 700 / 8 050 rec./s, one core
Overhead per record	104 B (64 sig. + 8 seq. + 32 hash)
Tampered rejected / false rejections	100/100, 0

C. Limitations

Single-step spikes carry little evidence (23–48 % of readings). Rain scaling is hard (14–25 % of readings): about 90 % of intervals are dry and only 773 attacked rain rows exist for training. A coordinated regional shift is not separable from real weather using in-situ data alone; this is where the remote-sensing input is required, through a cross-source check against ERA5-Land [2]. Grading the trained detector on the station suite and adding the cross-source reference are the main Phase 2 extensions. The 2024 attacks share their generator with training (with new placements, interpolated sizes and three withheld families), so the figures are strong evidence rather than an independent benchmark.

VIII. FEASIBILITY AND LOCAL EXECUTION

Table IV summarises cost. All rows were measured on the same 2-core x86-64 machine, with the shipped artifacts (outputs in `reports/detector-overhead.json` and `reports/crypto-benchmark.json`); latency depends on array shape and model, not on values, so the measurement needs no data download. No purchase, GPU, cloud service or paid API is required, and dependencies are hash-locked.

IX. REPRODUCIBILITY

The test suite (745 passed, 17 skipped; skipped tests need a local broker; log in `reports/test-results.txt`) includes no-look-ahead and label-leakage tests for the detector. The frozen model ships with SHA-256 hashes; one command reproduces Table II from the public data and another retrains from scratch. A jury guide documents installation, data download, every command and how to read each output, and a command-line tool lets the jury plant its own attack and inspect the detector's p-values and alerts. Detector latency and memory are reproduced with `python -m eval.detector_overhead`.

X. CONCLUSION

Signatures and plausibility checks are complementary: Layer 1 makes record-level attacks deterministic, and the calibrated detector raises an alert on about four out of five valid-signature attacks at a low false-alarm rate, including attack families it never saw, although it flags only about one in three manipulated readings and slow drifts are caught late. Layer 4 ensures that what the detector misses or flags does not silently reach the analyst. The remaining blind spot, coordinated regional bias, is exactly the case that requires combining in-situ data with remote-sensing references, which is the focus of Phase 2.

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